CDS 101 – Final Project Report

Mason Minotaurs - Yejun Hwang, Jonathan King, and Andres Rodriguez

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> Replace this template text with your own writing, keeping the headings and overall structure aligned with the **CDS 101 project rubric**.

1. Problem Definition

This project will analyze the relationship between tuition costs and graduation rates based on a dataset from the US Department of Education. Specifically, whether institutions with higher net prices tend to have higher graduation rates. Graduation rate is used as a proxy for student success, while net price represents the average cost students pay after financial aid. This will be beneficial to incoming college students who want to see if the higher cost is worth it. College graduates may also be interested in seeing if their college tuition was worth it, or not, in addition to their experience after graduation.

We expect that colleges with higher net prices will have higher graduation rates. This expectation is based on the idea that more expensive institutions often have more financial resources, provide better academic support services, admit more academically prepared students, and have lower student-to-faculty ratios.

However, the relationship may not be extremely strong. Net price alone does not capture all factors influencing graduation rates. Other important variables include admission selectivity, student socioeconomic background, institutional type (public vs private), and Academic preparedness of students.

There is also a consideration that a more expensive college may have better education, so the graduation rates may be lower than a college one might consider "easier"; this relies on how well the education at a college is, which this model wouldn't show.

2. Data Acquisition & Description

- Name and source of the dataset(s).

College.rds dataset from the instructor

- How you obtained the data (download link, API, etc.).

Canvas download from CDS 101

- Brief description of the variables / features (with units if relevant).

There are many variables about college information, but we will only focus on C150_4 # 4-year graduation rate NPT4_PUB # public schools NPT4_PRIV # private schools ADM_RATE # selectivity CONTROL # public vs private LOCALE # urban/rural SATVRMID, SATMTMID # academic preparedness

- Approximate size (rows, columns).

Rows: 7,058 Columns: 1,977

- Any ethical, privacy, or bias considerations (if applicable).

There may be cultural and societal factors that are not accounted for, especially because there are divisions among race and sex. We are also going into the project with the assumption higher tuition may cause lower graduaiton rate, given a better education is more expensive and difficult.

3. Data Cleaning & Preprocessing

Describe the steps you took to make the data usable, for example:

- Handling missing values (drop, impute, etc.).

```
We dropped the rows that weren't a graduation rate of 4 years/ 4 year institutions
```

- Fixing types (dates, numeric vs. categorical).

```
We dropped rows that did not contain any information about the graduation rates and net prices (C150_4,
```

- Removing outliers (if done).

```
We didn't remove any outliers becuse they are needed for analysis
```

- Creating new features (feature engineering).

```
A new feature, net_price, was created by combining private and public institution net price
```

- Any scaling or encoding.

```
No scaling was needed
```

```r
# Example pseudo-code for cleaning:
college_clean <- college %>%
   dplyr::filter(PREDDEG == 3) %>%
   dplyr::mutate(
     grad_rate = C150_4,
     net_price = dplyr::coalesce(NPT4_PUB, NPT4_PRIV)
   ) %>%
  dplyr::filter(
    !is.na(grad_rate),
```

```r
    !is.na(net_price)
  )
```

4. Exploratory Data Analysis (EDA)

This section maps directly to the **EDA** part of the rubric:

- Summary statistics
- Visualizations
- Interpretation connected to the research question

```r
# library(dplyr)
# summary(data_clean)
```

```r
# library(ggplot2)
# ggplot(data_clean, aes(x = some_numeric_variable)) +
#   geom_histogram(binwidth = 5, color = "white") +
#   labs(x = "Some Variable", y = "Count", title = "Distribution of Some Variable")
```

5. Visualization Quality and Storytelling

Use this section to satisfy the **Visualization Quality** rubric criterion:

- Explain why your plot types are appropriate.
- Comment on labels, legends, colors, and overall readability.
- Mention any steps you took to make plots accessible and interpretable.

6. Modeling Approach

Explain how you framed the problem and which models you chose:

- Type of task (regression, classification, etc.).
- Baseline model or heuristic, if used.
- Main model(s) chosen and why they are appropriate.

7. Model Implementation & Evaluation

This corresponds to the **Model Implementation & Evaluation** rubric criterion.

Describe:

- Features used.
- Data splitting strategy (train/test or cross-validation).
- Metrics used (accuracy, F1, RMSE, etc.).

- Tables/plots summarizing performance.
- A short interpretation of each metric/plot.

```r
# Example structure:
# set.seed(123)
# train_index <- sample(seq_len(nrow(data_clean)), size = 0.8 * nrow(data_clean))
# train <- data_clean[train_index, ]
# test  <- data_clean[-train_index, ]
#
# model <- glm(target ~ ., data = train, family = binomial)
# preds <- predict(model, newdata = test, type = "response")
# # compute metrics...
```

8. Conclusions & Recommendations

Summarize the key takeaways:

- Answer your original research question(s) directly.
- Highlight the most important patterns or relationships you found.
- Discuss limitations (data size, bias, missing variables, etc.).
- Suggest possible extensions or future work.

9. Code Quality & Reproducibility

Briefly document how someone else can reproduce your results:

- Which R scripts or Rmd files to run.
- Any required R packages.

```
## R version 4.5.1 (2025-06-13 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 10 x64 (build 19045)
##
## Matrix products: default
##   LAPACK version 3.12.1
##
## locale:
## [1] LC_COLLATE=English_United States.utf8
## [2] LC_CTYPE=English_United States.utf8
## [3] LC_MONETARY=English_United States.utf8
## [4] LC_NUMERIC=C
## [5] LC_TIME=English_United States.utf8
##
## time zone: America/New_York
## tzcode source: internal
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
```

```
##  [1] broom_1.0.12    modelr_0.1.11   lubridate_1.9.4 forcats_1.0.1  
##  [5] stringr_1.6.0   dplyr_1.1.4     purrr_1.2.1     readr_2.1.6    
##  [9] tidyr_1.3.2     tibble_3.3.1    ggplot2_4.0.1   tidyverse_2.0.0
## 
## loaded via a namespace (and not attached):
##  [1] gtable_0.3.6       compiler_4.5.1     tidyselect_1.2.1   scales_1.4.0      
##  [5] yaml_2.3.12        fastmap_1.2.0      R6_2.6.1           generics_0.1.4    
##  [9] knitr_1.51         backports_1.5.0    pillar_1.11.1      RColorBrewer_1.1-3
## [13] tzdb_0.5.0         rlang_1.1.7        stringi_1.8.7      xfun_0.56         
## [17] S7_0.2.1           otel_0.2.0         timechange_0.4.0   cli_3.6.5         
## [21] withr_3.0.2        magrittr_2.0.4     digest_0.6.39      grid_4.5.1        
## [25] rstudioapi_0.18.0  hms_1.1.4          lifecycle_1.0.5    vctrs_0.7.0       
## [29] evaluate_1.0.5     glue_1.8.0         farver_2.1.2       rmarkdown_2.30    
## [33] tools_4.5.1        pkgconfig_2.0.3    htmltools_0.5.9
```

10. References

List any references you used, such as:

- Dataset documentation.
- Research papers or articles.
- Tutorials or blog posts.

Appendix (Optional)

Include extra plots, diagnostic checks, or model comparisons if needed.